

LAST NAME: _____

Practice Quiz 11B — MATH 2055 — Fall 2026

Practice for Quiz 11

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Auto-generated by the Codex harness using GPT-5.

State the strong inductive hypothesis in full and cover both base cases.

1. Let $a_1 = 4$, $a_2 = 7$, and $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 3$. Prove that $a_n = 3n + 1$ for every integer $n \geq 1$.